Hamza El Byad, Ph.D.

Machine Learning Engineer | Applied AI | Scientific Machine Learning

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Professional Summary

Machine Learning Engineer with a Ph.D. in Mathematics & Computer Science and experience building industrial ML pipelines, RAG-assisted applications and simulation-based optimization tools. Strong foundation in statistical time-series analysis, model validation, signal processing and scientific research.

Professional Experience

Rombo AI **Sep 2025 – Aug 2026**
Machine Learning Engineer | Cagliari, Italy

- Developed NMR spectral-processing pipelines and PLS/PCR regression workflows to predict crude-oil properties; used group-aware validation to prevent leakage across repeated scans.
- Worked on predictive modeling for fouling detection and industrial process monitoring.
- Contributed to a RAG-based internal assistant with structured catalog retrieval, conversational drafts, deterministic cost calculations and validation safeguards.
- Developed a SUMO/TraCI traffic-control simulator comparing fixed-time, Max Pressure and reinforcement-learning policies, with interactive performance visualization.

University of Cagliari **2022 – 2025**
Doctoral Researcher | Cagliari, Italy

- Performed statistical time-series analysis of high-energy X-ray observations, applying Fourier-based methods and uncertainty-aware inference.
- Analyzed 39 NuSTAR observations of M82 (2014–2024) to characterize QPO variability; published first-author research in The Astrophysical Journal (2025).
- Investigated regularized inverse modeling and resampling methods for detecting and interpreting weak signals.

European Space Agency (ESAC) **Jan – Jul 2023**
Data Science Research Intern | Madrid, Spain

- Applied deep-learning approaches to compact-source detection in Planck sky maps.

Astronomical Observatory of Cagliari (INAF-OAC) **Dec 2020 – Jul 2021**
Research Intern | Cagliari, Italy

- Developed Python tools for X-ray time-series feature extraction and statistical variability studies.

Selected Engineering Projects

NMR-Based Industrial Predictive Modeling | **Case study**
Built reproducible spectral preprocessing and interpretable regression workflows; evaluated unseen physical samples with group-aware validation.

RAG-Assisted Catalog Retrieval | **Case study**
Contributed to structured retrieval, stateful selection and internal estimated-cost safeguards; public case study only, company source kept private.

Adaptive Traffic Signal Optimization | **Case study**
Developed simulation-based controller comparison and visualization with SUMO/TraCI; no claimed real-world deployment.

Technical Skills

ML & Evaluation	scikit-learn, PyTorch, PLS/PCR, regression, group-aware cross-validation, feature engineering
Data & Signals	Python, NumPy, pandas, SciPy, spectral preprocessing, FFT, normalization, time-series analysis
Applied AI	RAG, embeddings, FAISS, LangGraph, Chainlit, contextual retrieval, LLM applications
Backend & MLOps	FastAPI, Git, GitHub Actions, automated testing, CI/CD, Docker, Linux
Simulation	SUMO, TraCI, reinforcement learning, Max Pressure control, optimization
Research Methods	Statistical inference, uncertainty estimation, signal processing, scientific computing

Education

Ph.D. in Mathematics & Computer Science **2022 – 2025**
University of Cagliari, Italy

M.Sc. in Applied Mathematics **2019 – 2021**
University of Cagliari, Italy

B.Sc. in Applied Mathematics **2015 – 2018**
Mohammed V University, Morocco

Selected Peer-Reviewed Publications

- El Byad, H., et al. (2025). “A NuSTAR Study of Quasiperiodic Oscillations from the Ultraluminous X-Ray Sources in M82.” Astrophysical Journal, 989(2), 202. First author. DOI
- Lai, E. V., et al. (2026). “High performance Stingray.” Astronomy and Computing, Article 101096. Co-author. DOI
- Bachetti, M., et al. (2023). “The High Energy X-ray Probe (HEX-P): Studying Extreme Accretion with Ultraluminous X-Ray Sources.” Frontiers in Astronomy and Space Sciences, 10, 1289432. Co-author. DOI

Selected Research Presentations & Training

- CNOC XII (2022): oral presentation; FERO-10 (2022) and CNOC XIII (2024): posters on X-ray variability and QPOs in M82.
- Doctoral training in astrostatistics (Milan, 2023) and inverse problems (Bonn, 2022).

Selected Certifications

- Machine Learning Foundations - University of Washington (Coursera, 2023).
- Fundamentals of Quantitative Modeling - University of Pennsylvania (Coursera, 2025).

Languages

English | Italian | French